MIDTERM PRESENTATION · OCTOBER 22, 2026

GYM APP

Nutrition, training and weight in one place — with suggestions that explain why progress stalls

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AGENDA

TEN MINUTES, FIVE PARTS

01	Business market study	Size, retention, competitors, segment
02	Business Model Canvas	The nine building blocks
03	Literature review & the gap	Six studies, one gap
04	Problem definition	What the project solves
05	Components & Gantt chart	What is built, what is next

01 · MARKET STUDY

A MARKET THAT DOUBLES BY 2030

$12.1B
Global fitness-app market, 2024

$25.8B
Projected market size, 2030

13.5%
Compound annual growth, 2024–2030

Growth is driven by **subscription apps** and **at-home and self-guided training** — exactly the space a personal tracker competes in.

Source: ResearchAndMarkets, “Fitness Apps — Global Strategic Business Report 2025–2030” (Dec 2025)

01 · MARKET STUDY

THE REAL PROBLEM IS RETENTION

3–8%

of health & fitness app users are still active on **day 30**

AppsFlyer: 3% · Adjust: 8% (27% on day 1)

45.7%

of health-app users had **stopped using** some apps

Main reasons: data-entry burden, loss of interest, hidden costs

Sources: Business of Apps, Health & Fitness App Benchmarks (2023) · Krebs & Duncan, JMIR mHealth uHealth 3(4):e101 (2015)

01 · MARKET STUDY

COMPETITORS TRACK STREAMS IN SILOS

App	Focus	Price (US)	Nutrition	Training	Cross-stream "why"
MyFitnessPal	Calorie & macro log	~$79.99/yr	Yes	Basic	No
MacroFactor	Adaptive diet coach	$71.99/yr	Yes	Separate app	No
Strong	Workout log	$29.99/yr	No	Yes	No
Hevy	Social workout log	$23.99/yr	No	Yes	No
Gym App	All streams, one engine	Free pilot	Yes	Yes	Yes, with evidence

To combine diet, training and weight today, a user pays for two apps and does the correlation in their head.

Annual list prices for the paid tier, US, from vendor and App Store listings, Oct 2026; regional prices vary.

01 · MARKET STUDY

TARGET SEGMENT AND POSITIONING

Who

- Recreational lifters, beginner to intermediate
- Actively cutting or bulking, so they already watch food and weight
- Pilot users: family and friends in Canada and the Middle East

Their pain

- Two or three apps to log one day
- Daily weight swings read as failure
- Stalled lifts with no idea why

For lifters who track food and training, **Gym App** is the tracker that explains *why* progress stalls — from data they already log, in under 15 seconds a day.

02 · BUSINESS MODEL CANVAS

HOW GYM APP CREATES AND CAPTURES VALUE

KEY PARTNERS

Supabase — auth, database, photo storage

Railway — API hosting

Cloudflare — web hosting

free-exercise-db — 876 exercises

KEY ACTIVITIES

Building the suggestions engine

Keeping logging under 15 s

Pilot testing

KEY RESOURCES

The analytics engine

Codebase, 335 tests

Users' own log data

VALUE PROPOSITION

Food, training and weight in one app

Any log done in under 15 seconds

Suggestions that show their evidence

Maintenance calories measured, not guessed

RELATIONSHIPS

Self-service

Weekly recap card

Rolling streaks — never reset to 0

CHANNELS

Web app, any device

Word of mouth

Shareable weekly recap

CUSTOMER SEGMENTS

Recreational lifters who are cutting or bulking

Beginner to intermediate

Pilot: family & friends, Canada and Middle East

COST STRUCTURE

API hosting ~$5/month · Supabase and Cloudflare free tiers · domain · development time

REVENUE STREAMS

Pilot is free · planned freemium: free logging, paid insights at [$__/month]

SELF-MONITORING WORKS

Burke, Wang & Sevick (2011) · J Am Diet Assoc

Systematic review, 22 studies

Self-monitoring diet, exercise and weight was consistently associated with weight loss — though study quality was weak.

Zheng et al. (2015) · Obesity

Systematic review, 17 prospective studies

Regular self-weighing went with greater weight loss and no adverse psychological effects.

For us: daily weight and daily food are the right inputs to ask for.

BUT PEOPLE STOP LOGGING

Krebs & Duncan (2015) · JMIR mHealth uHealth

National survey, 1,604 US adults

45.7% of app users had abandoned health apps — citing data-entry burden, loss of interest and hidden costs.

Butryn et al. (2020) · Obes Sci Pract

Digital self-monitoring in a weight-loss program

Adherence to eating and weight tracking fell significantly over time; more eating-tracking predicted more weight loss.

For us: ask for as little as possible, and derive everything else.

DIET, TRAINING AND WEIGHT INTERACT

Murphy & Koehler (2022) · Scand J Med Sci Sports

Meta-analysis, training in an energy deficit

A calorie deficit blunted lean-mass gains from resistance training; the bigger the deficit, the bigger the effect.

Hall et al. (2011) · The Lancet

Modelling energy imbalance and body weight

The static “3,500 kcal per pound” rule overestimates weight change; expenditure adapts, so it must be estimated dynamically.

For us: read a training stall alongside intake, and measure maintenance from the user’s own trend.

03 · THE GAP

WHAT RESEARCH SAYS VS. WHAT APPS DO

The literature shows	Current apps	Gap
Tracking helps only while people keep doing it	Ask for many inputs; logging fatigue	Minimal, high-signal inputs
Deficits change training outcomes	Diet and training in separate apps	One model across streams
Maintenance must be estimated from real data	Calculator guesses from height and activity	TDEE measured per user
Feedback must be specific to act on	Generic praise and badges	Suggestions with evidence

No current app uses the **combination** of a user's diet, training and weight to explain their progress — while keeping logging light enough to last.

How can a fitness app turn a minimal set of daily inputs — weight, macros and quick workout logs — into evidence-backed explanations of a user's progress, so that they keep logging and make better training and nutrition decisions?

01 · Under 15 s

Every logging interaction

02 · Derive, don't ask

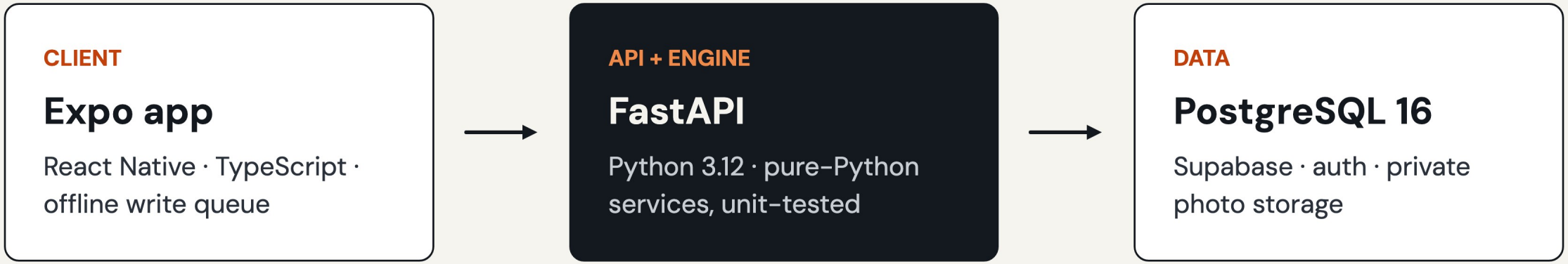
TDEE, trend, PRs, projections

03 · Explain why

≤ 3 suggestions, each with its numbers

05 · PROJECT COMPONENTS

SYSTEM ARCHITECTURE



Hosting

Cloudflare Pages · Railway · Supabase, co-located in N. Virginia

Security

ES256 JWT via public keys · row-level security on every table

Quality

335 tests · 99% engine coverage · CI on every push

05 · PROJECT COMPONENTS

FIVE SCREENS, EIGHT ENGINE SERVICES

THE APP

Dashboard | Workout | Nutrition | Weight | Photos

Weight smoothing
7-day trend hides water swings

Measured TDEE
Intake + 21-day weight slope

Goal projection
Least-squares date, refused if off-trend

Personal records
Epley estimated 1-rep max

Plateau detection
Stalls read with diet and rest days

Suggestions
At most 3, each with its evidence

Streaks & volume
Rolling 14 days; weekly sets per muscle

Weekly recap
Shareable summary card

Highlighted: the cross-stream core. Nothing derived is stored — every number is recomputed from the logs.

05 · WHAT THE USER SEES

SUGGESTION · PLATEAU

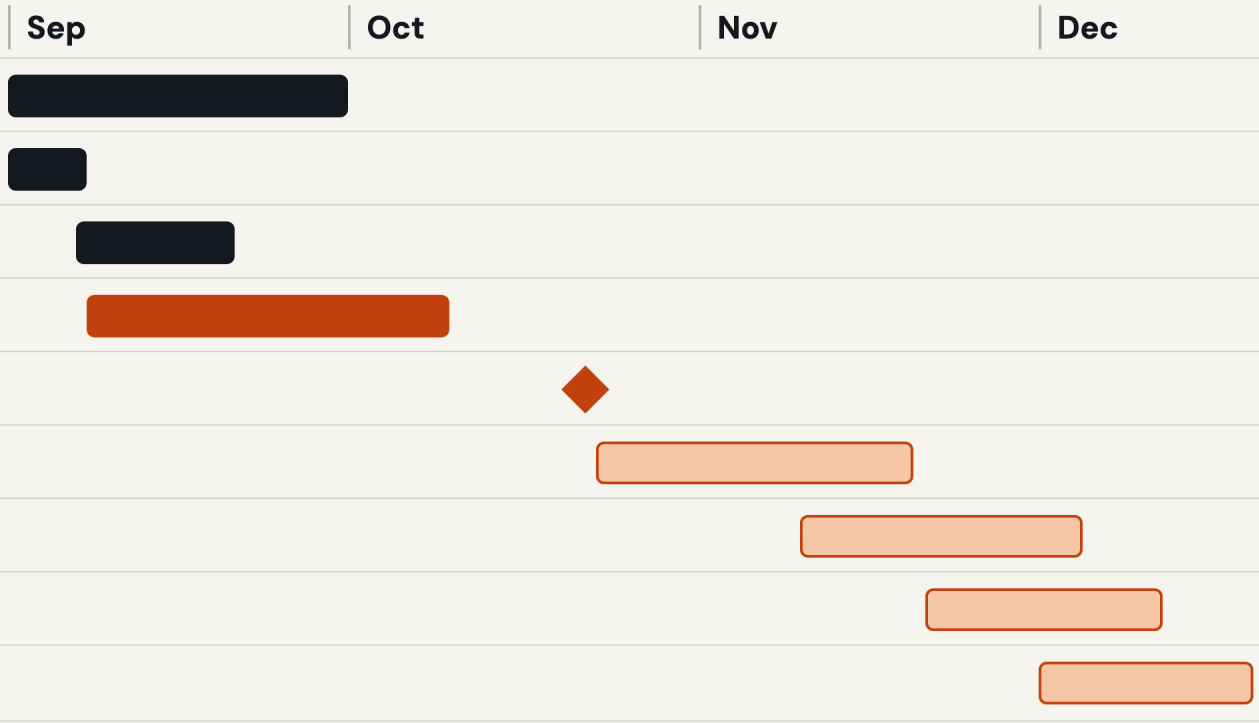
Bench stalled 4 sessions. Your average intake is ~400 kcal under maintenance over the same period — likely diet-related, not a training problem.

Two data streams, one sentence, numbers attached. No generic praise anywhere in the app.

05 · GANTT CHART

TIMELINE, SEPTEMBER – DECEMBER 2026

TASK	STATUS
Market study & lit. review	Done
Spec & architecture	Done
Build: phases 0–4	Done
Deploy & hardening	**Ongoing**
Midterm presentation	**Oct 22**
Pilot with real users	Planned
Evaluation & analysis	Planned
Iteration on feedback	Planned
Final report & defense	Planned



Dark = done · solid orange = ongoing · light orange = planned. Final deadline assumed mid-December — [confirm date].

SUMMARY

A growing market that loses 9 in 10 users within a month. Gym App keeps logging light and uses the combination of diet, training and weight to explain progress.

QUESTIONS?

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https://github.com/itsbkh99/Senior-Project-.git